



Restoring Punctuation in Uzbek Texts Using LLM's Fine Tuning Approaches

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Abstract

This study draws attention to the use of transformation-based ad-hoc tuning models for detecting and recovering punctuation marks in Uzbek language texts. The research was conducted to accurately predict punctuation marks including commas, full stops, exclamation marks and question marks using Bert and XML-Roberta architectures. The research is to develop a specialized database on key aspects of Uzbek punctuation, which helps the existence of focused research in this area. It has yielded good results in tests, confirming the potential of transformer models for punctuation recovery in low-resource languages, including Uzbek. The paper focuses on the development of tokenizers specialized exclusively for Uzbek. Methods for recovering punctuation marks in Uzbek language and improving model accuracy in case of class label mismatch were also discussed for future research. The study indicated that the mean F1 score in determining whether a word after 4 punctuation marks and punctuation marks is uppercase or lowercase in Uzbek language was 87.9%.

Keywords

Punctuation, Statistical methods, Machine Learning Algorithms, Neural Networks, RNN, BLSTM, Transformer models

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1 Introduction

Today there is a sharp development of artificial intelligence and its implementation in the form of deep neural networks based on modern architectures CNN, RNN, Transformer in various areas of human activity [1,2,3,4,5,6,7]. In recent years, significant progress has been made in natural language processing (NLP) algorithms and various language-related tasks, including punctuation recovery. Punctuation plays an important role in improving reading, comprehension and general linguistic diversity in written texts, but in many languages, including Uzbek, the absence or incorrect use of punctuation marks is a problem for humans. and poses a serious problem for automated language processing systems. Solving this problem requires the development of robust models capable of correctly recovering punctuation marks in texts written in languages with complex linguistic structures, such as Uzbek. This paper presents a new approach that uses fine-tuning models based on specially adapted transformers to restore punctuation marks in Uzbek texts.

Transformers have emerged as a powerful architecture in NLP and have demonstrated excellent performance on a wide range of language tasks. By customizing pre-trained transformer models to language-specific data, their contextual understanding and semantic representation capabilities can be used to effectively predict and recover punctuation occurrences in Uzbek texts. The use of transformer models is ideal for restoring punctuation marks. For example, in a study by Liu et al., transformer-based models show excellent accuracy and efficiency in punctuation recovery in English, French, and German. Their ability to identify semantic relationships allows them to accurately predict the placement of punctuation marks in different linguistic contexts. In addition, the authors show that transformer models outperform traditional rule-based approaches and other neural network architectures. This shows that transformer-based models are suitable for punctuation recovery tasks [8]. Overall, empirical evidence supports the feasibility and effectiveness of transformer models in punctuation recovery.

Punctuation is an element that determines meaning, clarity and harmony in the written language. In natural language processing, incorrect or missing punctuation marks in text are a huge problem that affects both human comprehension and automated processing tasks. Using semantic representation architecture and context understanding rules, sentences with missing or incorrectly placed punctuation marks can be reconstructed.

This study reviews the effectiveness of certain special adaptation models based on transformer technology used to reconstruct punctuation marks in Uzbek texts, taking into account the peculiarities of the language and its grammar.

This study focuses on analyzing the effectiveness of special adaptation models based on transformer technology used to reconstruct punctuation marks in Uzbek texts, taking into account its individual linguistic characteristics and punctuation rules. The effectiveness of the models is evaluated on data samples and compared with existing methods. It is also demonstrated that punctuation marks are useful for improving the accuracy of reconstruction and strengthening the skill of reading text. This research is aimed at helping the development of natural speech processing for understudied languages, including Uzbek. This study contributes to the solution of urgent problems of punctuation marks restoration in Uzbek speech. Special tuning of transformer models is required for punctuation marks restoration to adapt to the specific characteristics of the language, the field of application and statistics of information. This helps to increase accuracy, balances the distribution of information, eliminates inaccuracies and improves the efficiency of work.

Previously, various methods for punctuation marks restoration in the text were considered in this direction. Traditional approaches based on linguistic rules and experiments were studied [9], but, unfortunately, these methods rarely cope with the difficulties of the Uzbek language, where it is necessary to characterize complex grammar and limited linguistic resources.

Recently, solutions to punctuation restoration problems have found their highly effective solution based on neural networks. For example, Zhang and other researchers proposed a certain recurrent neural network (RNN) model with attention mechanisms. It showed a distinctive result in punctuation prediction [10]. In a similar way, Liu et al. presented a transformer-based architecture for punctuation restoration. This architecture showed the ability to handle different linguistic structures. The field has evolved in specific settings with the help of the advances in transformers. In 2018, Devlin et al. presented the BERT (Bidirectional encoder Representations from Transformers) model. This model was later adapted for many natural language processing tasks. It is worth noting that BERT shows high performance in punctuation restoration, which is confirmed by the study of Brown and colleagues [11]. In addition, models such as GPT (Generative Pre-trained Transformer) show excellent results in extracting contextual dependencies, which are key to correct punctuation [12]. Recently, research has focused on specific languages and datasets for punctuation restoration. For example, Uygur Kurt and his team apply specially adapted transformer models such as BERT, ELECTRA, and ConvBERT on a newly created Turkish dataset for this task. In their results, due to the balanced class distribution, the Precision, Recall, and F1 scores are presented, where the ELECTRA Transformer model achieved 83.9% in the overall F1 score. Also in their work, Tran and colleagues

presented a new adaptive Transformer-based model for the prediction task. The model is designed to efficiently label Vietnamese sentences with minimal computational and memory consumption. Due to the powerful capabilities of the Transformer architecture, the proposed model showed high performance in solving problems related to the structure and punctuation of the Vietnamese language due to the ability to capture long-range dependencies in sequential data [13].

Furthermore, Tanvirul A. and colleagues propose a strategy to solve the punctuation restoration problem using Transformers by considering both high-resource (e.g., English) and low-resource (e.g., Bangla) languages. It is interesting to note that the XLM-RoBERTa model (in the large version) achieved an overall F1 score of 87.0% on this test dataset [14]. These studies provide valuable insights into the generality and adaptability of punctuation repair methods in different language contexts.

Overall, the studies analyzed demonstrate ongoing efforts to implement best practices in punctuation repair using both traditional methods and neural network-based approaches.

1.1 Description of Restoring Uzbek texts Punctuation methodology

This section discusses the methodology used to restore punctuation marks in Uzbek texts using transformer models. The paper begins with a description of the dataset used for training and evaluation. In this context, information about the content, size, and interpretation process of the dataset is provided, which is necessary to understand the training context.

Then, the transformer models used in the study, such as the BERT and XLM-RoBERTa architectures, and the process of tuning them are discussed. The choice of hyperparameters, optimizer, and training duration are also discussed. This section presents the reflections and methodological decisions made when adapting transformer models to the task of restoring punctuation marks in Uzbek texts.

1.1.1 Features of the Dataset specification. Any deep learning model requires a dataset for training, and this dataset serves as the main source of the deep learning model. One of the problems encountered during the research was the lack of ready-made open-source datasets for tuning special punctuation recovery models for the Uzbek language. Therefore, there was a need to create a data set during the research process. The task of punctuation recovery can be directly related to the problem of classifying tokens in text. Punctuation marks are treated as separate lexemes in the text, such as period (.), comma (,), exclamation mark (!), question mark (?). Spaces () can be treated as separate characters if there are no visible characters.

The sequence of characters in the text follows a certain rule, which allows you to calculate the probability that punctuation marks will appear in certain places. Models based on transformers, consisting of encoder and decoder blocks with a special mechanism, play an important role in solving this task. Research work begins with the creation of a training data set. Previous studies have shown that training datasets for punctuation retrieval tasks can be automatically generated from correct punctuation marks in the target language, eliminating the need for manual labor [14, 15]. To

Table 1: Dataset characteristics.

Sets	Percentage (%)	Sentences	Words	Unrepeated words
Train set	78	16 890 234	75 854 069	62 093
Validation set	12	2 598 497	11 669 856	9 552
Test set	10	21 654 14	9 724 880	7 960
Total	100	21 654 145	97 248 805	79 605

Table 2: Token distribution.

Sets	Total	OU	OO	.O	!O	,O	?O	.U	!U	,U	?U
Train	75 854 069	75 854	42 661 962	12 160 968	608 048	13 965 373	3 378 046	1 638 447	682 686	409 611	273 074
Val	11 669 856	13 671	6 224 340	1 690 918	97 545	2 648 522	532 699	251 044	104 088	61 014	46 012
Test	9 724 880	9 920	5 514 601	1 759 024	75 754	1 290 432	343 082	240 057	86 542	54 214	351 254
Total	97 248 805	99 445 (0.1%)	54 400 903 (56%)	15 610 910 (16%)	10781 347 (0.8%)	17 904 327 (18%)	4 253 827 (4%)	2 129 548 (2%)	873 316 (0.8%)	524 839 (0.5%)	670 340 (0.6%)

do this, it is important to collect texts with correct punctuation in the Uzbek language.

Research shows that larger data sets increase the accuracy of custom tuning models. Therefore, as part of the effort to develop language models for a continuous Uzbek speech recognition system, a text corpus consisting of approximately 15 million sentences and a total of 80 million words was created. Among these sentences, approximately 68,000 are not repeated [16]. In another similar work, about 2 million sentences were used to create a speech corpus for training a neural network model in the creation of speech analysis systems in Uzbek [16,17,28]. In addition, texts with correct punctuation were collected from official state websites (for example, lex.uz platform of legal documents [18]). The split ratio in sample training of the collected text corpus is shown in the following table:

In this work, the task of recovering four punctuation marks in the text is considered: these are period (.), comma (,), question mark (?), exclamation mark (!). In addition, in addition to the recovery of the four punctuation marks shown in the work, the study also covered the problem of determining whether the word following these punctuation marks is written in upper or lower case. According to the classification scheme, there are 10 classes of token classification. To facilitate analysis, punctuation marks in the text are replaced by predefined images. Here, "AB" represents the punctuation mark "A" in the specified position, and "B" represents the beginning of the word after it with an uppercase or lowercase letter.

The observation of these characters in the collected text sentences is shown in Table 2 below. In the table, "U" indicates an Uppercase letter, "O" if the token does not indicate any punctuation, or "O" if it is in an uppercase letter.

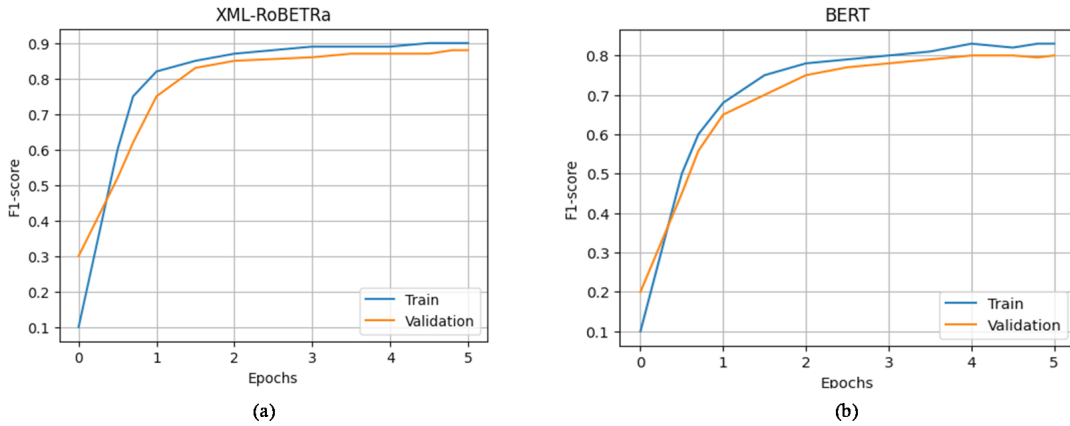
If each token is represented using the notation described above, and each word is given a unique numeric identifier, a suitable data set for transformer models can be prepared. Tokenization. This process, known as tokenization, involves breaking text into individual tokens or words and converting them into digital images. These digital images allow the transformer model to understand

and process text efficiently. This process is covered in the Hugging Face documentation. [19, 20]. After creating a dataset, you need to process the data before applying it to your model. One important step in this processing is vectorization, which involves converting raw text (a sequence of words) into a format suitable for machine learning algorithms. Vectorization maps each token in the text to its corresponding index in the dictionary, allowing the model to process text data as digital input. There are different approaches to creating vocabularies for tokenization, each with its own advantages. As discussed by Hugging Face users and other literature, these approaches typically include word-level tokenization, character-level tokenization, and subword tokenization. In the study, subword tokenization was chosen because it is effective in working with morphologically rich languages such as Uzbek.

While researching the pre-trained multilingual tokenizers available in the Hugging Face repository, it became clear that some already support the Uzbek language. However, when these tokenizers are applied to Uzbek text, vocabulary is generated when used in the teaching process, which may be related to the limited amount of information available in Uzbek. To overcome this limitation, a tokenizer specially designed for the Uzbek language text was developed. BertTokenizer and the XML RoBERTa tokenizer were selected for our tokenizer models, both of which perform space tokenization. It involves creating a new language-specific dictionary file by configuring the tokenizer. In the study, a "vocab.json" file with a vocabulary size of 50,000 tokens was created using SentencePiece Trainer [21]. Additionally, special characters such as "PAD", "UNK", "CLS", and "SEP" are introduced as part of this dictionary to control the operations encountered during tokenization. In the process of creating the working tokenizer, we used the unigram model, which is a statistical model widely used in part-word tokenization tasks. This model was effective in covering the complex features of Uzbek language lexemes, and made it possible to lexeme tokens more accurately and efficiently.

Table 3: Example of vectorization using tokenization.

Input text	“Insonlar kitoblarsiz bugungi faravon kunga etib kelishmas edi.” (eng. People would not have reached today’s prosperity without books)
Tokens	['_Inson', 'lar', '_kitoblar', 'siz', '_bugungi', '_faravon', '_kunga', '_yetib', '_kelish', 'mas', '_edi', ':', '</s>'] (eng. '_People', 's', '_books', 'you', '_today', '_prosperity', '_today', '_arrive', '_come', 'mas', '_was', ':', '</s>')
Token ID	[7940, 47, 4006, 303, 2453, 41441, 10974, 3517, 5383, 6325, 69, 6, 50100]

**Figure 1: (a) and (b) graphs, respectively, show the variation in F1 score accuracy during training stages of BERT and RoBERTa models**

1.1.2 Selected models based on the transformer. The study evaluated the reliability of two transformer models. After various studies, it was found that the BERT and XML-RoBERTa models consistently gave positive results in many studies. Therefore, these models were chosen for the research on punctuation restoration in Uzbek language.

BERT (Bidirectional encoder Representations from Transformers), XML-RoBERTa (Cross-lingual language model – RoBERTa) are all transformer-based models used for natural language processing (NLP) tasks, but they also have a number of key differences:

BERT: BERT uses a bidirectional transformer architecture. It processes all tokens of the input sequence by binary encoding [10].

XML-RoBERTa: XML-RoBERTa is an extension of RoBERTa, which is an improved model over BERT[22]. It uses a similar architecture to BERT, but uses a more optimized training method.

BERT and XML-RoBERT have a maximum input sequence length, typically around 512 tokens. They perform efficient processing of longer sequences. BERT was pre-trained primarily on English textual data and initially focused on English-language tasks. However, it is customized and customized for different languages. XML-RoBERTa is a multilingual model trained on text data in multiple languages. It aims to provide robust performance across languages without the need for language-specific models. In general, BERT and XML-RoBERT are powerful transformer models widely used for NLP tasks [10,15,23,24,25].

1.1.3 Settings for customizing the model for the Uzbek language.

In this research, special tuning of pre-prepared BERT and XML-RoBERTa models was performed. The dataset created for the Uzbek language has been modified again by changing certain settings in the experiments.

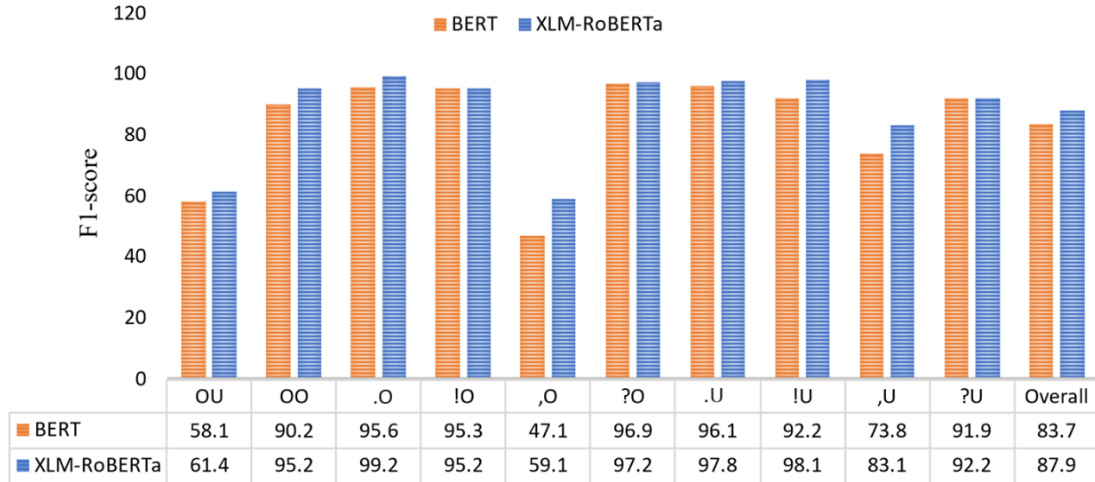
During the customization process, several basic settings were used to optimize the learning process. These settings include a sequence length of 256, a set size of 64, a learning rate of $2e-5$, and a weight decay of 0.01. In addition, the training scheme covered five epochs to ensure comprehensive improvement of the model.

An optimizer characterized by a learning rate of $5e-4$, an optimizer beta of 0.9 and 0.999, and an epsilon (eps) value of $2e-05$ was used for optimization [26]. Utilizing the computational power of the NVIDIA V100S - 32 GB CPU for training and inference tasks, it facilitated efficient model training and inference processes.

During training, the BERT and XML-RoBERTa models showed different durations for each epoch, 75142 seconds for the BERT model and 9278 seconds for the XML-RoBERTa model, respectively. In order to monitor the progress and performance of the training process, the average F1 score values are calculated for each 1000 steps for the training and test datasets, and reports are given on the recall and accuracy of the model. The training results for the considered models, including their performance over different training periods and datasets, are visually depicted in Figure 1 above, providing a complete overview of the training and performance trends of the models. represents the appearance.

Table 4: Average accuracy indicators of models

Models	Tokenizers	Train			Validation			Test		
		P	R	F1	P	R	F1	P	R	F1
BERT	Pretrained BertTokenizer	70.3	72.8	71.5	70.3	71.5	70.8	68.9	70.3	69.6
XLM-RoBERTa	XML-Roberta Tokenizer	75.2	74.4	74.8	77.8	76.1	76.9	78.1	73.5	75.7
BERT	Proposed Bert Tokenizer	81.4	82.1	81.7	83.5	82.7	83.1	82.4	85.1	83.7
XLM-RoBERTa	Proposed XLM-RoBERTa Tokenizer	89.6	90.3	89.9	89.6	87.6	88.5	88.6	87.2	87.9

**Figure 2: Average values of F1 score accuracy by class for fine-tuning models.**

1.1.4 Evaluation indicators. Class Disparity is a new set of Uzbek punctuation data. Therefore, job evaluation indicators should be selected accordingly. Three indicators were used for this: precision - precision, recall and F1 scores for each label. These indicators are defined as follows:

$$Precision (P) = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 - Score = \frac{2 * Precision * Recall}{Precision + Recall} = \frac{2 * TP}{2TP + FP + NP}$$

TP represents the number of true true positives, F1 represents the number of false positives, and FN represents the number of false negatives. Precision (P), recall (R) and F scores (F1) are calculated taking into account the unbalanced class distribution.

Therefore, the use of these indicators is considered a suitable assessment for the problem [27]. Attempts were made to train models using pre-built tokenizers and custom tokenizers. The average accuracy of the models in different parts of the dataset is shown in Table 4 below.

Table 4 clearly shows how the proposed tokenizer XML-Roberta for the Uzbek language and the developed tokenizer XML-RoBERTa sharply fire its high accuracy and the accuracy of 87.9% for the test set with the same metric F1-score. The diagram presented in Figure 2 below describes the accuracy of the character prescription

in percentage (%) for the two models Bert and XLM-RoBERT in the class and based on the F1-score, which shows that the proposed model is most likely to be wrong in cases where the display is closed.

1.1.5 Restrictions. This research focuses on the main four types of punctuation, although all 10 were taken into account in the set of classes. In addition, the study is limited to the use of individual solutions such as the BERT model and its followers. In future studies, it is planned to expand the number of punctuation marks for a wider coverage of the goal, thereby achieving more effective prediction of punctuation marks in Uzbek texts.

2 Conclusion

This scientific article describes a method for targeted punctuation restoration in Uzbek texts based on deep neural network architectures, which was then trained using a specially formed dataset of punctuation in Uzbek contexts. The trained models were tested using two well-known transform models, such as BERT and XML-RoBERTa. The testing results were analyzed in terms of their ability to predict the presence of four basic punctuation marks: commas, periods, exclamation marks, and question marks. The analysis of the experiments showed impressive accuracy figures for determining 4 selected punctuation marks and amounted to 87%. In conclusion, the results of the study help and are a small worthy step in the development and solution of text classification problems from the

field of natural language processing for languages such as Uzbek, and pave the way for the creation of more complex and accurate punctuation restoration systems in the future.

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